

Math727 – Numerical Analysis II

Homework 3

1. Given a SPD matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, show that

(a) For the CG method the following representations for $n = 1, \dots, n_*$ hold:

$$\mathbf{x}_n = q_n(\mathbf{A})\mathbf{b} \quad \text{for a suitable } q_n \in \mathbb{P}_{n-1},$$

and

$$\mathbf{r}_n = -p_n(\mathbf{A})\mathbf{b} \quad \text{with } p_n(t) = 1 - tq_n(t)$$

(b) It holds that $q_n(t) = \sum_{j=0}^{n-1} c_j t^j$ in which the vector $(c_0, \dots, c_{n-1})^T \in \mathbb{R}^n$ is the solution of the following linear system of equations:

$$\begin{bmatrix} \mathbf{b}^T \mathbf{A} \mathbf{b} & \mathbf{b}^T \mathbf{A}^2 \mathbf{b} & \dots & \mathbf{b}^T \mathbf{A}^n \mathbf{b} \\ \mathbf{b}^T \mathbf{A}^2 \mathbf{b} & \mathbf{b}^T \mathbf{A}^3 \mathbf{b} & \dots & \mathbf{b}^T \mathbf{A}^{n+1} \mathbf{b} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{b}^T \mathbf{A}^n \mathbf{b} & \mathbf{b}^T \mathbf{A}^{n+1} \mathbf{b} & \dots & \mathbf{b}^T \mathbf{A}^{2n-1} \mathbf{b} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{bmatrix} = \begin{bmatrix} \mathbf{b}^T \mathbf{b} \\ \mathbf{b}^T \mathbf{A} \mathbf{b} \\ \vdots \\ \mathbf{b}^T \mathbf{A}^{n-1} \mathbf{b} \end{bmatrix}$$

2. Prove the following three-terms recursive relation for the residual in the CG method:

$$\mathbf{A} \mathbf{r}_k = -\frac{1}{\alpha_k} \mathbf{r}_{k+1} + \left(\frac{1}{\alpha_k} - \frac{\beta_{k-1}}{\alpha_{k-1}} \right) \mathbf{r}_k + \frac{\beta_{k-1}}{\alpha_{k-1}} \mathbf{r}_{k-1}.$$

3. Consider a matrix of form $\mathbf{A} = \mathbf{I} + \alpha \mathbf{B}$ where $\mathbf{B}$ is real skew symmetric (i.e., $\mathbf{B} = -\mathbf{B}^T$).

- a. show that $(\mathbf{A} \mathbf{x}, \mathbf{x}) / (\mathbf{x}, \mathbf{x}) = 1$ for all nonzero $\mathbf{x}$.
- b. show that the resulting Hessenberg matrix by the Arnoldi process will have the following tridiagonal form:

$$\mathbf{H}_k = \begin{pmatrix} 1 & -\eta_2 & & & \\ \eta_2 & 1 & -\eta_3 & & \\ & & \ddots & & \\ & & \eta_{k-1} & 1 & -\eta_k \\ & & & \eta_k & 1 \\ & & & & \eta_{k+1} \end{pmatrix}.$$

4. Let a matrix $\mathbf{A}$ have the form

$$\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{Y} \\ \mathbf{0} & \mathbf{S} \end{pmatrix}.$$

Assume that GMRES method is used to solve a linear system with the coefficient matrix $\mathbf{A}$. Let

$$\mathbf{r}_0 = \begin{pmatrix} \mathbf{r}_0^{(1)} \\ \mathbf{r}_0^{(2)} \end{pmatrix}$$

be the initial residual vector. It is assumed that the degree of the minimal polynomial of $\mathbf{r}_0^{(2)}$ with respect to $\mathbf{S}$ (i.e., the nonzero polynomial p of lowest degree with $p(\mathbf{A})\mathbf{r}_0^{(2)} = \mathbf{0}$ and the coefficient of its highest order term being 1.) is k . What is the maximum number of steps that GMRES would require to converge for this system? (Hint: Consider the sum $\sum_{i=0}^k \beta_i (\mathbf{A}^{i+1} - \mathbf{A}^i) \mathbf{r}_0$ where $\sum_{i=0}^k \beta_i t^i$ is minimal polynomial of $\mathbf{r}_0^{(2)}$ with respect to $\mathbf{S}$.)

5. **(Programming Exercise)** Let $\Omega = (0,1)^2$. Consider numerical solution of the boundary value problem

$$\begin{cases} -u_{xx}(x, y) - u_{yy}(x, y) + a1(x, y)u_x(x, y) \\ \quad + a2(x, y)u_y(x, y) + b(x, y)u(x, y) = f(x, y), & (x, y) \in \Omega \\ u(x, y) = 0, & (x, y) \in \partial\Omega \end{cases} \quad (1)$$

where $a1(x, y) = x$, $a2(x, y) = y$ and $b(x, y) = 1 + x^2 + y^2$. We let $u(x, y) = xy(1 - x)(1 - y)$ and then the right hand side f is determined correspondingly. We consider an uniform square grid with grid points (x_i, y_j) which are chosen to be equidistant:

$$x_i = ih, \quad y_j = jh, \quad i, j = 0, 1, \dots, N \quad (h = 1/N)$$

and the center difference scheme:

$$\begin{aligned} & \frac{-U_{i-1,j} - U_{i,j-1} + 4U_{i,j} - U_{i+1,j} - U_{i,j+1}}{h^2} \\ & + a1_{i,j} \frac{U_{i+1,j} - U_{i-1,j}}{2h} + a2_{i,j} \frac{U_{i,j+1} - U_{i,j-1}}{2h} + b_{i,j} U_{i,j} = f_{i,j}, \\ & i, j = 1, 2, \dots, N-1, \end{aligned}$$

where $a1_{i,j} = a1(x_i, y_j)$, $a2_{i,j} = a2(x_i, y_j)$, $b_{i,j} = b(x_i, y_j)$ and $f_{i,j} = f(x_i, y_j)$, and

$$U_{i,j} \approx u(x_i, y_j), \quad i, j = 1, 2, \dots, N-1$$

$$U_{i,0} = U_{i,N} = U_{0,i} = U_{N,i} = 0, \quad i, j = 0, 1, \dots, N.$$

Let $\mathbf{x}$ be the vector consisting of unknowns $U_{i,j}$ ($i, j = 1, 2, \dots, N-1$) ordered row-wise (from left to right) and upward. Then, we get a linear system such as

$$\mathbf{Ax} = \mathbf{b}$$

where $\mathbf{A} \in \mathbb{R}^{(N-1)^2 \times (N-1)^2}$ is a sparse matrix and $\mathbf{b} \in \mathbb{R}^{(N-1)^2}$. For $N = 10, 20, 40$, determine an approximate solution of (1) using the GMRES method, where the iteration is stopped when either more than $Kmax = 100$ iterations are reached or the residual $\|\mathbf{r}_k\|_2 < \epsilon \|\mathbf{b}\|_2$ with $\epsilon = \frac{h^2}{100}$. You may use the Matlab code in the lecture notes. Provide the number of required iteration steps k , and the errors $\max_{i,j=1,2,\dots,N-1} |U_{i,j} - u(x_i, y_j)|$ in a table. Also plot the history diagram of number of iterations k vs. the relative residual $\|\mathbf{r}_k\|_2 / \|\mathbf{b}\|_2$ in a semi-log scale for the $N = 40$ case using Matlab (the command *semilogy*).